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Multiple Pose Virtual Try-On Based on 3D Clothing Reconstruction

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ABSTRACT Recently, image-based virtual try-on (VTON) systems using deep generative models have drawn significant research attention. However, the 2D clothing shape transform methods in the earlier works show serious limitations in 3D clothing deformation required in multiple-pose VTON scenarios. In this paper, we develop a 3D-MPVTON system of two pipelines and show that a 3D clothing model reconstruction approach provides much better results for the multi-pose VTON scenario. First, the 3D clothing model reconstruction pipeline is based on CloTH-VTON+. The try-on clothing is matched to the target clothing regions in the simply-shaped reference human model and its 3D model is reconstructed using the associated 3D human body model. For natural clothing rendering from an arbitrary view, accurate texture mapping is crucial. We have developed a highly accurate texture matching method. The try-on pipeline first generates target segmentation from the target pose for conditional information for the following stage's network models. Correct target segmentation was one of the main performance bottlenecks in previous VTON studies. Our proposed equalized entropy loss for the target segmentation generation network greatly reduces the segmentation label imbalance, and results in high-quality segmentation and reduced training time. The rigged reconstructed 3D clothing model can be easily deformed into the target pose and the human body shape while retaining the textures of the clothes. The remaining parts, i.e., the non-target clothing regions of the human in the target pose are generated through a deep generative human pose transfer model. Finally, the generated clothing and the remaining parts are combined using conditional generative networks to in-paint the dis-occluded areas and blend them together. Our proposed hybrid pipeline outperforms the previous 2D-based approaches by substantial margins in both objective evaluation and subjective user study, especially in the cases of the large pose and viewpoint changes.

INDEX TERMS Virtual try-on, 3D clothing reconstruction, arbitrary poses, pose transfer, hybrid approach.

I. INTRODUCTION

Online shopping and e-commerce are rapidly increasing worldwide due to their efficiency and convenience. However, in the case of fashion items, especially clothing, a fair percentage of purchased products are returned due to a mismatch in style and size. Virtual Try-On (VTON) technology was invented to narrow the gap between these online viewing and purchasing experiences by creating virtual images that help users have a more thorough and realistic shopping experience.

Early approaches with 3D computer graphics could not handle mass-market demands due to the high costs and

difficulty of building 3D clothing and human models. Recently, with advances in deep learning technology, several methods such as VITON [1], CP-VTON [2], CP-VTON+ [3], ACGPN [4], CloTH-VTON [5], CloTH-VTON+ [6] have been proposed. They create virtual fitted images based on the store clothes but in a fixed input body pose, which we will call here 'Fixed Pose VTON (FPVTON)'.

However, customers often have a desire to see themselves in the clothing from different perspectives or in different poses. According to a user study in FashionOn [7], 78% of users are more willing to use try-on services with multiple views and different poses than those with only one fixed posture. Multiple pose VTON (MPVTON) systems can generate virtual fitted images in multiple poses with different try-on clothes from a single customer image

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as input. These MP-VTON methods can be built by combining clothing warping techniques developed in FPVTON and human pose transfer techniques. Due to their practical importance, techniques for human pose transfer have been widely studied, [8]–[15]. Most of the state-of-the-art (SOTA) clothing warping methods from the fixed pose VTON study [1]–[4], [16]–[19] use 2D transformation algorithms, e.g., Thin-Plate-Spline (TPS) transformation [20] or, more recently, optical flow-based methods [11] for the try-on clothing deformation. However, they often fail to generate good clothing deformation results, especially when the try-on clothes have long-sleeves and the reference human has a complex 3D pose. The inherent limitation of these 2D deformation algorithms is that they cannot deform the try-on clothing into real 3D postures [5], [6].

It is crucial to realize that the clothing fitted on the human body experiences 3D deformations from joint motion of human limbs which can not be approximated by any 2D non-rigid transformation like 2D ARAP (as-rigid-as-possible) and TPS (thin plate spline). For typical examples, 1) the sleeve covers the body part of clothing, i.e., self-occlusion or folding and 2) the displacement direction of cloth on the human body part is neither smooth nor correlated. Also, in the multi-pose scenarios, the viewpoint can be from the side or the back as well as a front view. We define a 'simple' target pose as the front view pose close to the A-pose, where the displacement from the in-shop clothing image to the target image can be approximated by 2D non-rigid transforms, i.e., no folding or self-occlusion due to the pose change and no occlusion or dis-occlusion of clothing due to viewpoint change. We will show and discuss a few challenging pose and viewpoint cases together with simple cases in V.

The existing pipelines often hide these artifacts in deformation in the final try-on output by utilizing the in-painting properties of the deep networks [2]–[4]. These approaches work partially for mono-colored in-shop clothes but exhibit serious limitations for logo-ed or patterned clothing [3], [5], [6], [21].

Recently, CloTH-VTON [5] and CloTH-VTON+ [6] show that a 3D clothing model can be reconstructed from a simple human 3D body model, which can then generate a natural 3D deformation, which overcomes the clothing deformation problem for large spatial transformations in FP-VTON cases. In this paper, we extend this 3D clothing reconstruction-based hybrid approach to multiple pose virtual try-on cases.

The proposed method, named 3D-MPVTON, can be divided into two main pipelines, 3D clothing model reconstruction and try-on synthesis. The 3D clothing model reconstruction pipeline follows the approach proposed in Cloth-VTON+ [6]. First, the try-on clothing is matched to the target clothing regions in a simple-shaped reference human model, and the clothing 3D model is constructed by referencing the matching elements in the associated 3D human body model. For natural clothing rendering from an arbitrary view, highly accurate texture mapping is crucial,

and we have developed just such a highly accurate texture matching method. The try-on pipeline consists of four stages following the approach proposed in MG-VTON [19]. Target human semantic segmentation (THSS) generates a target segmentation image. Correct target segmentation is crucial because it forms the principal conditional information for the generative networks in the following stages. We propose two key modifications to the original THSS networks which significantly improve the target segmentation results and training time by applying both improvements together. First, we use more informative masks to identify all the regions except the clothing to be replaced. Second, a balanced loss function is used to take the imbalance in the labels into account. The rigged reconstructed 3D clothing model can be easily deformed to the target pose and the human body shape while retaining the textures of the clothes. The remaining parts, i.e., the non-target clothing regions of the human in the target pose, are generated through a deep generative human pose transfer model. Finally, the generated clothing and the remaining parts are combined utilizing conditional generative networks to in-paint the dis-occluded areas and blend them all together. Then, a coarse try-on generation network fuses all these elements to generate the coarse try-on results. However, these coarse results are generated with blurry textures, and so a refinement network is utilized to generate the final try-on results.

The sample results are displayed in Fig. 1. The quality improvement using 3D reconstruction and deformation becomes significant in multiple pose scenarios compared to those with fixed poses. Creating new perspectives requires that the system be able to color and texture angles that are not seen in the original image. When clothes are worn on the body, structures such as logos, shapes, and text must be as natural as possible. The deformations of the clothing must be consistent with the body and posture of the reference model.

The contributions of the paper are summarized as follows:

- We extend CloTH-VTON+ to a multiple-pose scenario and show that 3D clothing reconstruction can provide higher quality VTON results than the previous multi-pose guided VTON systems. The quality improvement over these systems is substantial because our approach is able to handle the significant deformations of the in-shop clothing needed to handle a large range of target poses.
- We propose a new fast high-quality loss and training method for the target human semantic segmentation generation network compared to that in MG-VTON. Notably, the technique takes the label imbalance in the dataset into account. Efficient generation of high-quality target human semantic segmentation plays a crucial role in guiding the following stages.
- We propose a new clothing texture mapping method to reduce the artifacts generated at the revealed boundaries of clothing, which is critical for high-quality results, especially for side view poses.

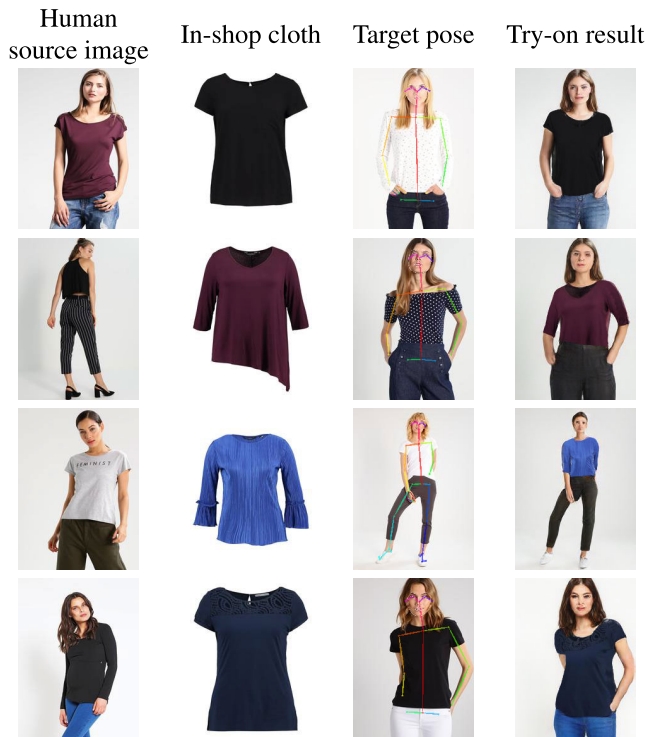


FIGURE 1. Examples of multiple-pose virtual try-on images generated by 3D-MPVTON. From left to right: source human images, new in-shop clothes, target poses (joints) and the virtual try-on results. The final try-on images take skin, hairstyle, and lower-body clothes from the source human images and replace the upper-body clothes with new in-shop clothes in the target poses.

II. RELATED WORKS

Multiple-pose virtual try-on (MP-VTON) is a complex system consisting of several multi-stage pipelines. It uses a number of different image analysis and generation algorithms, many from well-established prior works. Reviewing all of these constituent technologies is beyond this paper's scope, so we discuss here only those closely related to the current application.

A. IMAGE-BASED FIXED-POSE VIRTUAL TRY-ON

The first generation image-based VTON systems change the clothing of the customer's image with a new in-shop clothing image while keeping the same pose and viewpoint as the original ones. Most of the previous systems consist of two stages: the first stage for warping the in-shop clothing to the target area in the human image (resize, align, transform), and the second stage for blending the remaining source regions with the warped clothing segment, as well as generating disclosed parts. For the blending stage, most methods use U-Net [22] style networks to synthesize try-on on a human reference image. For the clothing warping stage, variations of 2D non-rigid deformation are used. VITON [1] generates coarse warped clothing first and then warps the in-shop clothing images using Shape Context Matching (SCM) [23]. CP-VTON [2], CP-VTON+ [3] and ACGPN [4] regress the TPS parameters using convolution neural network-based geometry matching [24], [25]. ClothFlow [11] uses finer

optical flow-based deformation through deep learning-based correspondence matching. Detailed study [6] shows these 2D warping methods have limited capabilities, especially when the target pose has a 3D pose difference from the in-shop clothing.

B. IMAGE-BASED MULTIPLE POSE VIRTUAL TRY-ON

Subsequent to the fixed-pose image-based VTON efforts, researchers began to generate the try-on image from the customer images with new clothing and arbitrary postures. [26] generates a new body shape before generating warped clothing and applies bidirectional GAN to generate final try-on results. FashionON [7] tries to generate newly synthesized human segmentation from given input new clothing and target pose and then generates blurry results, and finally improves the details on the face by using a refinement network to get a more realistic image. The state-of-the-art MG-VTON [19] also uses synthesizing segmentation to help the appearance network generate natural try-on results. However, these methods use 2D warping [24] to generate the warped clothes, which often fails to produce realistically deformed clothes, especially in cases of long-sleeved clothing or difficult new target poses [5], [6]. Unlike MG-VTON, we use a 3D body model to reconstruct 3D clothing and are thereby able to generate more realistic deformed clothing textures.

C. 3D HUMAN MODELS AND RECONSTRUCTION

Modeling a human body is a valuable technique for building human virtual reality applications, including virtual try-on. Recent efforts build 3D human body models based on parametric statistical models, such as SMPL [27] and SMPL-X [28]. Several parameter estimation methods for SMPL model have been proposed using joint and silhouette input, SMPLify [29], SMPLify-X [28], HMR [30], SPIN [31], ExPose [32]. However, the estimation accuracy is not yet good enough. The effects of this limitation will be discussed in the corresponding method and experiment section.

The modeling and reconstruction of clothed humans have been well-studied and continues to remain challenging. Photo Wake-up [33] uses SMPL human body models to reconstruct the clothed human model from a single human image and then animates the model. BodyNet [34] and Pifu [35] use a deep learning network to generate volumetric human models.

However, generalized and complete reconstruction of the human model is still challenging due to the large range of human poses and shapes and the resulting occlusion and deformation present in the source images. A more tractable approach first fits clothing to a standard body model and then generates reconstructions by manipulating and rendering that fitted model in 3D. The recent CloTH-VTON [5] and CloTH-VTON+ [6] systems leverage this fact and propose that this 3D reconstruction method be applied to VTON applications. This paper extends this approach to multiple-pose scenarios. Zhou *et al.*'s work [36] is most similar to our approach. However, they use mannequin-fitted

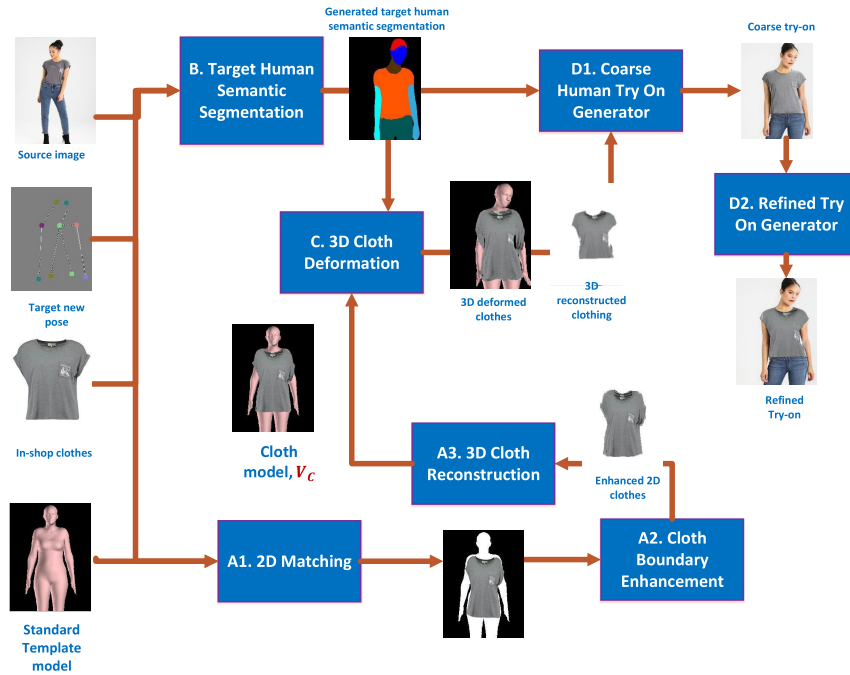


FIGURE 2. The proposed 3D-MPVTON pipelines A1, A2, and A3 are the components of 3D clothing reconstruction pipeline III-A. The try-on pipeline is composed of 4 stages. B. Target human semantic segmentation III-B generates Generated target human semantic segmentation. 3D clothing is deformed to target pose in C. 3D Clothing Deformation III-C. A coarse result is generated in D1. Coarse Human Try On Generator III-D1. And D2. Refined Try-On Generator III-D2 generates refined try-on results.

clothing images rather than in-shop clothing images and did not extend their study to virtual try-on applications.

III. 3D-MPVTON

Fig. 2 shows two pipelines of our 3D-MPVTON. The clothing reconstruction pipeline described in III-A is formulated as

$$C_{3D} = F_{3DREC}(C, M_C, SMPL), \quad (1)$$

where C , denotes an item of in-shop clothing, M_C its clothes mask, and $SMPL$ the template SMPL body model. Standard (A)-posed 3D clothing models are prepared through a 3D clothes reconstruction process before the customer's VTON request, similar to CloTH-VTON+ [6]. The try-on pipeline takes an input tuple consisting of a human source image I_{src} , an in-shop clothing image of the try-on clothes C and a target pose P_t , and generates the final try-on image I_{ref} whose human likeness comes from I_{src} including the face, skin color, hair, and lower clothes, but is shown wearing the in-shop clothing C in the target pose P_t . We assume the input image is processed for source human segmentation M_{src} , pose P_{src} , and 3D clothing model C_{3D} corresponding to user's in-shop clothing choice C in advance.

$$I_{ref} = F_{3D-MPVTON}(I_{src}, M_{src}, P_{src}, C, M_C, C_{3D}, P_t) \quad (2)$$

The try-on pipeline comprises 4 stages. B. Target human semantic segmentation III-B generates the target human semantic segmentation. 3D clothing is deformed to target pose in C. 3D Clothing Deformation III-C. A coarse result is generated in D1. Coarse Human Try-On Generator III-D1.

And D2. Refined Try-On Generator III-D2 generates final try-on results.

One can consider reusing the existing FPVTON and human pose transfer in a cascade pipeline for the MPVTON problem. This cascade approach was examined in a previous work, FashionOn [7], where they compare their results with 'Pose Transfer + CP-VTON' and report that the results of Pose transfer and FPVTON have artifacts or lack details, especially for human limb occlusion, duplicating erroneous pleats and shadows, and limiting dramatic posture transformation. In our opinion, applying processing in a cascade pipeline degrades the output image quality due to the loss of original inputs. FPVTON and Pose transfer use similar processing. In our method, we combine the two stages into a more compact one (in Fig. 2). First, FPVTON and Pose transfer share the same target human semantic segmentation. In FPVTON, we need to generate warped clothing through A1, A2, A3, and C in Fig. 2, after which the try-on module in FPVTON and pose transfer is combined into D1. Because of this combination, the whole pipeline is more compact than the separate pipelines of FPVTON and pose transfer for MPVTON. Moreover, by putting all the inputs of warped clothing and original source image into "D1. Coarse Human Try-On Generator", D1 can learn to generate better new pose try-ons.

A. 3D CLOTHING RECONSTRUCTION

Our 3D clothing model reconstruction is composed of geometry reconstruction and texture image mapping. For the

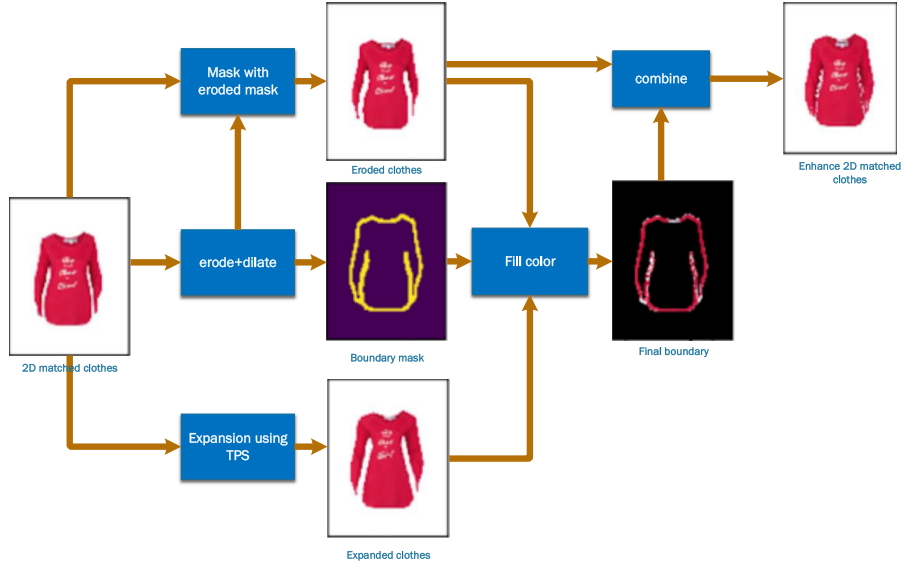


FIGURE 3. Clothing texture matching procedure. From the 2D-matched clothes and their masks, we apply erosion and dilation to get an extended boundary mask. By applying TPS [20] to get the expanded clothing region. Then we combine the boundary region with the eroded clothing texture to get the extended 2D matched clothing texture.

geometry reconstruction from 2D in-shop clothes, we adopt the method described in CloTH-VTON+ [6]. We briefly summarize the method here.

A standard human body model V_S is defined as an A-posed SMPL [27] with a fixed pose and shape parameter set. We first generate the target human segmentation S_S^m using the trained THSS generator as described in the Sec. III-B.

$$S_S^m = G_{THSS}(M_{sh}, M_{sf}, M_{sbs}, M_{sl}, M_{ss}, I_{sbot}, C, P_t) \quad (3)$$

where C is in-shop clothing, M_{sh} , M_{sf} , M_{sbs} , M_{sl} , M_{ss} , I_{sbot} , P_t are the hair mask, face mask, body shape mask, leg masks, mask shoe masks, lower-body clothing mask and the pose of the template SMPL A-Pose model. The 2D matched clothing I_C^m is generated by applying SCM-TPS [23] to the try-on clothing according to the matching silhouette mask M_C^m which is simply the upper clothing labels from S_S^m . According to the study in CloTH-VTON+ [6], SCM-TPS achieves satisfactory deformation results due to the minor 3D pose difference between the A-pose of the standard SMPL body model and the in-shop clothing.

Once the matching in 2D image space is done, the depth of the clothing model is constructed. Assuming similarity in the depth between clothes and a human body, the depth of the 3D body model is utilized. First, the standard 3D body vertices V_S are projected into 2D image space. Then, a second 2D matching is applied to the 2D projected vertices of the standard model to align with the clothing boundary. Transformed 2D mesh points are considered as the vertices projected from the 3D mesh of clothing V_C . Mapping from 2D points to 3D points is done with inverse projection using depth obtained from the body with a small constant gap to produce the reconstructed 3D clothing model.

$$V_C = P^{-1} \cdot (F_{SCM-TPS}(P \cdot V_S), z(V_S)) \quad (4)$$

Here, P is the projection matrix with camera parameters $K \cdot [R|t]$, P^{-1} is the inverse projection matrix of the same camera, and $z(V_S)$ is the distance measured from the camera to the vertices.

2D matched try-on clothing images are used as the texture UV mapping for the 3D clothing mesh. For natural and visually accurate rendering at arbitrary poses and views, the clothing texture mapping step is crucial. In CloTH-VTON+, the warped clothes on the template body model $C_{3D-warped}$ contain artifacts such as those shown in the Fig. 6 (a) on the boundary of each clothing. These artifacts come from the imperfect 2D matching of I_C^m with 3D clothing models, mainly due to minor inaccuracies in the clothing mask M_C^i and errors in 2D matching. To minimize these artifacts, we developed a new texture enhancement and mapping algorithm to address this problem, especially on the boundaries of I_C^m . To deal with the imprecision in the clothing mask, we apply morphological operations on I_C^m as shown in Fig. 3 so that the UV-texture projection area can be completely filled with the clothing texture.

On the mask M_{cm} of I_C^m , we apply erosion and dilation operations to get an eroded mask M_e and a dilated mask M_d , respectively. We employ masking of M_e on I_C^m to get the eroded clothes I_e . We combine M_e and M_d to get the boundary mask $M_b = M_e \oplus M_d$. We estimate contour points on M_{cm} and M_d . For each contour point on M_d , setting overlapping the two contour points in the same plane, we search for the nearest contour point on M_{cm} to obtain a pair of corresponding points, one point on M_{cm} and the other on M_d . From those pairs of corresponding points, we estimate the TPS [20] transformation parameters and use these to warp I_C^m to get the boundary-expanded clothes I_x . Then, we paint the boundary mask with the color from I_C^m , expanded clothes I_x , and a shifted version of I_x to get the final boundary. Then, the final

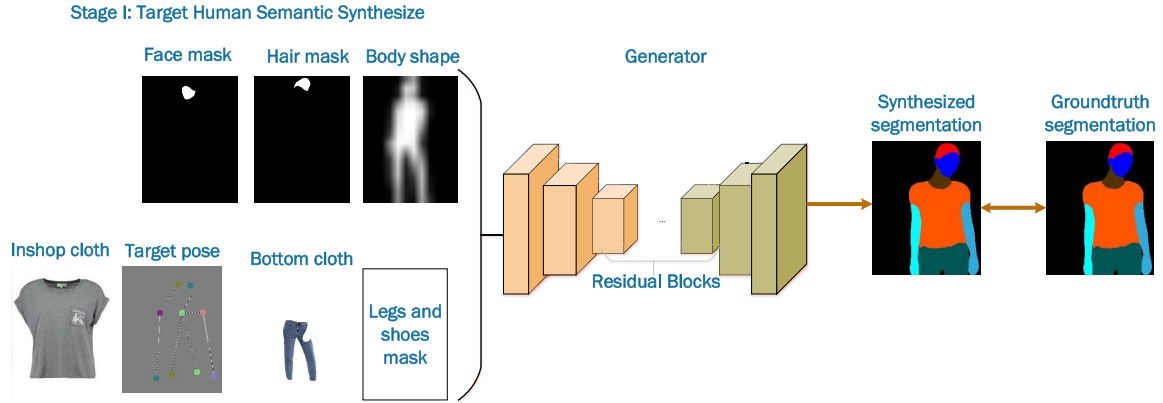


FIGURE 4. Target human semantic segmentation generator (THSS). THSS takes the source face mask, the source hair mask, the body shape, the in-shop cloth, the target new pose, the leg masks and the lower-body parts as the inputs, then generates the target human semantic segmentation.

boundary is combined with the I_e to get the final enhanced 2D matched clothing.

B. TARGET HUMAN SEMANTIC SEGMENTATION GENERATION

Earlier human pose transfer networks generate the output image directly using the target pose as the conditional input. More recent human pose transfer systems [4], [16], [19] use an approach in which a target semantic segment map is generated first and used for pixel generation guide and is shown to perform well. Target human segmentation is used as the guide information for all following processing stages in our pipelines.

The network inputs for the human image should be agnostic with respect to the current clothing since different shaped clothing is used at testing time. To make the input clothing-agnostic, MG-VTON uses the source hair mask M_{sh} , source face mask M_{sf} , source body shape M_{sbs} , but no other parts, especially the lower-body clothes area. (Here, “mask” means the area of the image that the network uses). Therefore, MG-VTON fails to recognize the lower part of the body correctly and estimates pants for the lower-body clothes area. Since only the upper clothes are changed, we can keep the network input agnostic of the new in-shop clothes by adding masks for the lower-body clothing from the source human image to the network’s input. Furthermore, we add masks for the source legs (M_{sl}), source shoes (M_{ss}) and source lower-body clothing (I_{sbot}). We do not combine all lower body parts into one mask because each input has a different role for guiding the THSS to generate the segmentation. The color on lower-body clothing provides information for the THSS to generate segmentation of lower clothing types, such as skirts and pants (or shorts/trousers), while legs and shoes guide the THSS to generate the same segmentation at the output of THSS.

$$S_{tgen} = G_{THSS}(M_{sh}, M_{sf}, M_{sbs}, M_{sl}, M_{ss}, I_{sbot}, C, P_t) \quad (5)$$

The reason why MG-VTON mostly generates pants (Fig. 5 a) is that, most humans in the training MPV dataset are



FIGURE 5. Comparison between target human semantic segmentation generated by MG-VTON [19] (a) and our 3D-MPVTON (b). From left to right: Human source image, the in-shop clothes, the target pose visualization, the generated target human semantic segmentation by MG-VTON (a), the target human semantic segmentation generated by the proposed THSS (b) and some color maps.

wearing pants. After training, Stage I of MG-VTON learned the statistical from the dataset to predict lower body parts as pants. With additional masks, THSS can generate more accurate output for lower-body parts (Fig. 5 b).

Fig. 4 shows the diagram of the THSS generator (G_{THSS}). We used the Pix2PixHD [37] for generator and discriminator networks.

The loss function of THSS networks is composed of L1 loss and GAN loss.

$$L_{THSS} = \arg \min_{G_{THSS}} \max_{D_{THSS}} (\lambda_{bce} * L_{THSS_{BCE}}^{G_{THSS}} + L_{THSS_{GAN}}) \quad (6)$$

The GAN loss can be as follow:

$$L_{THSS_{GAN}} = E_{S_{thss_{in}}, S_{tgt}} [\log D(S_{thss_{in}}, S_{tgt})] + E_{S_{thss_{in}}} [\log (1 - D(S_{thss_{in}}, S_{tgen}))] \quad (7)$$

where $S_{thss_{in}} = (M_{sh}, M_{sf}, M_{sbs}, M_{sl}, M_{ss}, I_{sbot}, C, P_t)$, and S_{tgt} is the target ground truth human semantic segmentation. The loss of segmentation label estimation is defined as

$$L_{THSS_{BCE}}^{G_{THSS}} = - \sum_{n_c} S_{tgt} \log(G_{THSS}(S_{thss_{in}})) + (1 - S_{tgt}) \log(1 - G_{THSS}(S_{thss_{in}})) \quad (8)$$

In the baseline network, small and infrequent labels such as shoes and legs require more training time and yet get poor estimation accuracy. To handle this problem, we apply



FIGURE 6. Comparison between 3D warped clothing generated by CloTH-VTON+ [6](upper row) and the proposed 3D-MPVTN(lower row). CloTH-VTON+ generates the 3D reconstructed clothing with artifacts on the boundary while our proposed method reduces these artifacts.

an ‘equalized’ label weighted cross-entropy loss, which uses weights so that all labels have the same contribution to the total loss. A statistical analysis of the MPV dataset has produced a histogram (H) of the segmentation labels. First, for each label i , we count the total number of pixels whose label is i for each image in the dataset follows Eq. 9. Then we normalize histogram values and get the weights by inverting the histogram as Eq. 10. Finally, these weights (W) are used for training the THSS with an “equalized” label weighted cross-entropy loss.

$$H_i = \sum_k \sum_x \sum_y (S_k(x, y) == i) \quad (9)$$

where H_i is the histogram of segmentation label i , S_k is the ground truth semantic segmentation of the image I_k , and $S_k(x, y)$ is the segmentation label of the pixel at the location (x, y) in the image I_k .

$$W_i = (\sum_i H_i) / H_i \quad (10)$$

After training with ‘equalized’ label weighted loss for 3 epochs, we switch back to the normal cross-entropy loss. The improvement gained with our methods compared to MG-VTON is shown in Fig. 5.

C. 3D CLOTHING DEFORMATION

We first deform the 3D clothed model obtained in the Sec. III-A according to the target pose and shape to get the target clothing model V_C^t , and then project it to 2D to get the 3D warped clothes image $C_{3D-warped}$.

$$C_{3D-warped} = F_{3DXfer}(C, M_C, P, V_S) \quad (11)$$

The comparison result with CloTH-VTON+ in Fig. 6 shows how the texture mapping refinement greatly reduces artifacts on the clothing boundary, especially when the target pose presents a side view of clothes.

D. SYNTHESIS OF TRY-ON IMAGE

1) COARSE HUMAN TRY-ON GENERATOR

To generate realistic final try-on results, recent systems such as CloTH-VTON+ [6], ACGPN [4] often use the Pix2PixHD multi-scale structure [37] in their generators and discriminators. In the Pix2PixHD network structure, the U-Net style

is used. Furthermore, U-Net requires that all inputs must be well-aligned to generate the output.

The procedure in Fig. 7 can be express as Eq. 12.

$$I_{coarse} = G_{CHT}(C_{3D-warped}, P_t, S_{tgen}, I_{snocloth}, SFT(S_{sgt}, S_{tgen})) \quad (12)$$

where Source Feature Transformer ($SFT(S_{sgt}, S_{tgen})$) which is called the pre-trained geometric matcher in MG-VTON [19] is the pre-trained model to estimate the transformation between S_{sgt} and S_{tgen} , $I_{snocloth}$ is the human source image where the top clothing area is painted with the color white.

Clothing areas including lower body parts, source hair, and skin, go first through the source branch of the ‘encoder’ in CHT to get source features. However, source features are not aligned with the target human body feature generated from the other branch of the CHT encoder. The spatial transformer, based on the correspondence between synthesized human segmentation and source segmentation, is used to align the source features to the target position. These transformed and aligned features are concatenated with clothing features, and then are decoded in the ‘decoder’ of CHT to generate a coarse try-on. Please note that in multi-pose try-on, the identity information, e.g., human face, skin, and hair, can not be preserved the same way as in fixed-posed VTON because it also experiences the change of pose and viewpoint. The identity information is transferred through CHT networks which deforms the input information and generates dis-occluded parts (generative network). The CHT network we borrowed from the MG-VTON loses detailed/local information that can cause some differences in face details.

MG-VTON uses a Warp-GAN to warp valuable source features to a target location using pre-trained networks whose inputs are the source and target semantic segmentation. However, a wrong target segmentation, especially in the lower body parts, causes incorrect feature matching. This prevents MG-VTON from being able to preserve the identity of the human source image well. With the improvement of the target segmentation in THSS and the improved warped clothes, our coarse VTON results outperform MG-VTON’s.

For training, MG-VTON uses synthesized human segmentation from THSS for the inputs. Noticing that we can use the segmentation output for the target in the training

Stage III: Coarse human try-on generator

Generator

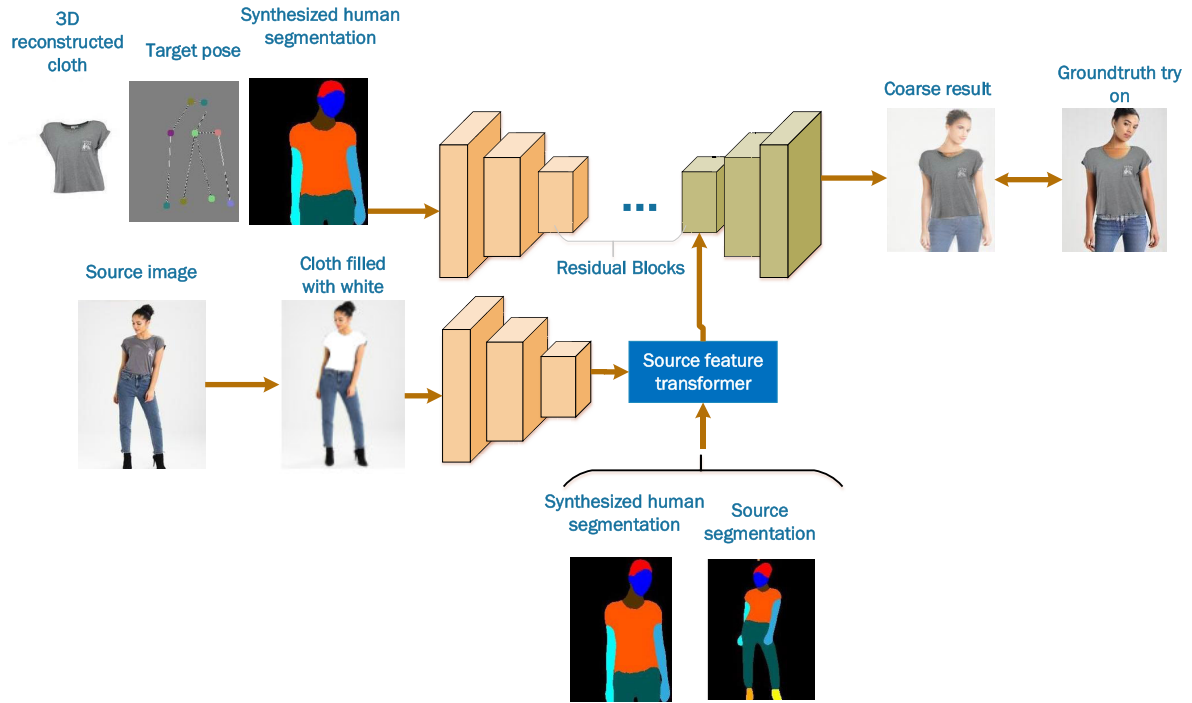


FIGURE 7. Coarse human try-on network diagram (CHT): CHT takes the 3D reconstructed clothing, the target human pose, the synthesized human segmentation, the source human image with top clothing painted in white color and the source feature transformer ($SFT(S_{tgt}, S_{tgen})$) as inputs and then generates a coarse try-on result.

Stage IV: Side awareness refinement module

Coarse result

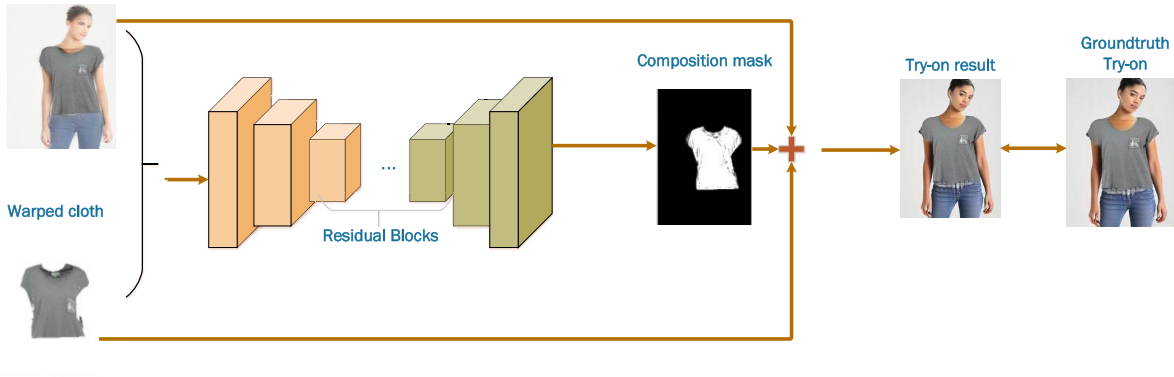


FIGURE 8. Refinement Try-On Network (RTO). RTO takes the 3D reconstructed clothing and the coarse try-on result as the inputs, and generates a composition mask which is used as the alpha mask for blending the 3D reconstructed clothes and the coarse try-on.

period, we replace synthesized human segmentation (S_{tgen}) with ground truth semantic segmentation (S_{tgt}). Using S_{tgt} , we guarantee that CHT receives the correct target human semantic segmentation to forward source identify to the target location. Then CHT tries to generate the correct ground-truth target image for calculating loss.

2) BACK-VIEW-AWARE TRY-ON REFINEMENT

CP-VTON [2], CP-VTON+ [3] and MG-VTON [19] use a refinement network to further enhance the texture of clothes

for the coarse try-on stage. Nevertheless, in the MPV dataset, all of the in-shop clothes are front-view, and so all the back-view clothing in the out is generated by the CHT network. Moreover, we do not need to copy the front-side view clothes texture to the back-side target pose try-on. Unlike MG-VTON, we use a ‘back-view aware strategy’ to enhance clothing texture in the coarse-stage results further. This back-view-aware strategy is based on the left, and right shoulder location of target pose P_t and detects the polarity of the target pose. It does not apply a refinement network and produces the

coarse try-on result as a generated try-on when the target pose is back-view.

$$I_{ref} = C_{3D-warped} * M_{com} + (1 - M_{com}) * I_{coarse} \quad (13)$$

where $M_{com} = G_{CHT}(C_{3D-warped}, I_{coarse})$ is the composition mask generated by the RTO generator G_{CHT} with inputs are 3D reconstruction cloth $C_{3D-warped}$ and coarse result from CHT I_{coarse} .

IV. EXPERIMENTS

A. DATA SET PREPARATION

We used the MPV dataset [19] for all of our experiments, which consists of 35,687 single-person images in a variety of poses, and 13,524 images of in-shop clothes. Each image has a fixed resolution of 256×192 . We split them into training and test sets of 52,236 and 10,544 three-tuples, respectively. We generate a further 10,544 three-tuples of different clothes for testing.

Since our clothing reconstruction method uses in-shop clothes and their masks, after checking the MPV dataset, we decided to filter out the seemingly undesirable in-shop clothing images. First, we filtered out images with two or more clothes, called a ‘clothing stack’ (Fig. 9). In total, 356 of the original 13,524 images were removed in this filtering. Also, we filtered out the incorrect clothing masks (See Fig. 9 for example).



FIGURE 9. Samples of bad clothing masks (upper row) and clothing stack (lower row).

B. IMPLEMENTATION DETAIL

Three neural networks in our approach, THSS, CHT and RTO, all share the U-Net architecture [22] for the generators, and the discriminators are from Pix2PixHD [37] (except RTO, which does not have a discriminator). GAN losses include the generator losses, discriminator losses for real and fake outputs, and the feature-matching loss [37] but only for CHT. All networks are implemented in PyTorch. THSS, CHT, and RTO are trained for 50, 20, 5 epochs, respectively, with a mini-batch size of 8. For testing, we use two sets of testing tuples. One contains the tuples that have the same in-shop clothes and different poses. The other set includes tuples with different clothes and different poses, similar to the real-world scenario used for visual comparison, e.g., as shown in the qualitative analysis and user study in this article. A learning rate of 0.0002 is used for training all the networks.

For optimization, ADAM optimizer with $\beta_1 = 0.5$, $\beta_2 = 0.999$ is used.

Shape context matching [23] is used to find the control points between the in-shop clothing mask and template body mask, then a thin plate spline [20] is used to transform in-shop clothes to fix the body silhouette template mask. Since the TPS does not perfectly match in-shop clothing to template body shape, we further perform some geometric adjustments on the 2D clothing matching to cover all the template body mask areas. From input in-shop clothes, we build a 3D clothing model from the 2D matching texture. Then the texture is transferred based on target pose and shape, which is estimated by SMPLify-X [28] optimization. We then project the 3D reconstructed clothes to 2D to get the deformed clothes image.

We built and trained the coarse human try-on and refinement networks following the MG-VTON. To understand the effects of the target segmentation performance, in addition to the try-on results using our THSS, we also show the results with MG-VTON’s original synthesized segmentation in quantitative and qualification.

V. RESULTS AND ANALYSES

In this section, we present the results of our method compared to the baseline, MG-VTON [19]. First, we show the performance improvement in target human area generation, both in accuracy and training speed. Then, we present quantitative, qualitative, and user study results for the final try-on images.

A. TARGET HUMAN SEMANTIC SEGMENTATION GENERATION

Table 1 compares the intersection-over-union (IOU) and pixel accuracy of the synthesized human segmentation from our method and MG-VTON, i.e., with and without lower-body parts as inputs and the use of equalized cross-entropy loss. The experimental results show significant improvement of our method over MG-VTON’s, especially for the regions which are small or rarely appear in the dataset and lower-body clothing area. IOU is calculated on each label separately for a precise evaluation of the methods. MG-VTON fails to predict lower-body parts such as skirt, leg, socks, dress, coat (only 10% accuracy), because there is no information in MG-VTON’s input to guide the network for predicting the lower-body parts. Hence, for the most cases, the lower-body parts are predicted to be pants. By adding lower-body part information, our methods can significantly improve the prediction accuracy for lower-body parts (increases of more than 50% for legs, 80% for the skirt, 10% for the socks, 58% for the coat, and 82% for the dress). Due to the effect of equalized weight loss, accuracy is also improved for small parts. Specifically, the prediction accuracy for head-wear increased from 0% to 33%, socks from 3% to 18%, and legs from 13% to 65%. Table 1 shows that our proposed method can generate excellent segmentation, especially for small label regions, and faster training time. There are no labels for gloves in the MPV dataset, so there are some “NA” values. Although some parts are extremely rare in the dataset, such as sunglasses and

TABLE 1. Comparison of target human semantic segmentation results. We provide a quantitative comparison of the generated target human semantic segmentation output here, between the baseline MG-VTON's segmentation generation network and THSS with our proposed loss and training method. Pixel accuracy and the intersection-over-union (IOU) metrics are used for the comparison.

Label	MG-VTON [19] (200 epochs)		The proposed (50 epochs)	
	Accuracy	IOU	Accuracy	IOU
background	0.978	0.953	0.985	0.964
hat	0.0	0.0	0.334	0.326
hair	0.722	0.615	0.771	0.691
glove	NA	NA	NA	NA
sunglasses	0.0	0.0	0.0	0.0
upper-clothes	0.917	0.789	0.934	0.877
dress	0.027	0.025	0.849	0.717
coat	0.014	0.013	0.590	0.546
socks	0.036	0.031	0.179	0.162
pants	0.855	0.662	0.907	0.832
neck	0.663	0.546	0.814	0.675
scarf	0.0	0.0	0.0	0.0
skirt	0.090	0.077	0.889	0.723
face	0.895	0.829	0.938	0.855
left-arm	0.786	0.678	0.870	0.775
right-arm	0.759	0.647	0.882	0.764
left-leg	0.139	0.115	0.679	0.581
right-leg	0.132	0.110	0.653	0.570
left-shoe	0.632	0.448	0.618	0.489
right-shoe	0.628	0.430	0.664	0.507

TABLE 2. Comparison between the original method in MG-VTON and the proposed method. Performance metrics, mean SSIM [38], MS-SSIM [39], FID [40], and LPIPS [41] are obtained on the different pose and same cloth cases in MPV [19] dataset to evaluate against the ground truth. For mean SSIM, MS-SSIM, a higher score means better results, while FID and LPIPS metrics, the lower values shows the better output.

Method	Mean SSIM [38]↑	Mean MS-SSIM [39]↑	FID [40]↓	LPIPS [41]↓
MG-VTON [19]	0.723	0.686	15.397	0.185
3D-MPVTON (with MG-VTON's Segmentation)	0.7319	0.711	18.360	0.179
3D-MPVTON (Ours)	0.7372	0.7299	13.715	0.168

scarves, both the two methods can not segmenting, these parts do not affect the performance much.

Due to the addition of equalized cross-entropy loss in our segmentation network during the training, the optimization time gets significantly reduced. For instance, MG-VTON trains their segmentation generation network for 200 epochs, whereas our THSS needs only 50 epochs for the training optimization.

B. VTON IMAGE RESULTS

1) QUANTITATIVE ANALYSIS

Using the target segmentation generated by MG-VTON [19], we further generate final try-on output by both MG-VTON and our method. We also generate the final try-on results from the proposed 3D-MPVTON with THSS's generated segmentation. We use several performance evaluation metrics, such as the Structural SIMilarity (SSIM) [38], Multi-Scale Structural SIMilarity (MS-SSIM) [39], Frechet Inception Distance (FID) [40] and Learned Perceptual Image Patch Similarity (LPIPS) [41]. Since these measures need ground truth for evaluation, we calculated these metrics on generated results from the testing set of the same clothing in different poses. Table 2 shows that our proposed method outperforms MG-VTON in all four metrics.

2) QUALITATIVE ANALYSIS

We examine the try-on results of 3D-MPVTON against MG-VTON's. We present both of the results for the different

poses and the same clothing scenarios in Fig. 10 and results for new pose and clothing scenarios in Fig. 11. In each case, warped clothing for MG-VTON and 3D reconstructed clothing for 3D-MPVTON are shown. The sixth column images in Fig. 10 and Fig. 11 are try-on results generated by the CHT generator with 2D warped clothing images in the fifth column. While the seventh and eighth columns (or tenth and eleventh columns) are 3D warped clothing and correspondent try-on with MGVTON segmentation (or with THSS output).

Typical try-on results with the same clothing in different poses are shown in Fig. 10. The short-sleeved clothing of the first, fourth, and sixth rows, MG-VTON, and our 3D-MPVTON show similar visual quality. The second row shows the case where clothing on the body was occluded by the right arm. For the side-view target pose of the third row, our 3D-MPVTON shows better quality than MG-VTON. First, the right sleeve should not appear after warping because of the occlusion of the right arm by the body, but it is still visible. Further, in the side-view, while changing the camera view from the clothing front face to the side-view, the bottom of the clothing will be smaller. The 2D deformation in MG-VTON tries to match the in-shop clothing in the front view to the target clothing area by wrongly narrowing the lower-body part. Our 3D reconstructed clothing hides the right sleeve, changing the point view of the camera and generating more naturally warped clothing. The fifth, seventh, and eighth rows show long-sleeve clothes and front-facing target human poses. In these cases, in-shop clothes are warped to the target



FIGURE 10. Qualitative comparison between our proposed method and MG-VTON on same clothes, different pose test set. From the left to right - Source human images, in-shop clothes, target new pose visualizations, human semantic segmentation generated by MG-VTON, 2D warped clothes generated by MG-VTON, the try-on results generated by MG-VTON, the 3D reconstructed clothes and the try-on results from our method based on MG-VTON's generated human semantic segmentation, the generated target human semantic segmentation from the proposed 3D-MPVTON, the 3D reconstructed clothes and the try-on results from the proposed 3D-MPVTON.

clothing segmentation without considering the occlusions and deformed textures. As a result of this, the deformed sleeves become significantly narrower in MG-VTON, especially in the seventh row, whereas our 3D warped clothing can hide the right sleeve of clothing, and the clothing is deformed naturally. Moreover, the texture on the warped clothing of MG-VTON gets enlarged unnaturally around the neck/chest region, while 3D-MPVTON preserves the realistic shapes and textures. Despite the texture matching algorithm, the artifacts at the clothing side/boundary for some results are noticeable.

Fig. 11 shows a selection of the try-on results from the test set of different clothing and different poses. MG-VTON, using 2D warping, also works in 'simple' poses or for short sleeved clothing which is not strongly affected by human poses (sixth rows). For non-simple poses with long sleeved clothing, MG-VTON fails to generate realistically-deformed clothing; the texture and logos can get heavily distorted (second, fourth, fifth, and eighth rows). On the contrary, 3D-MPVTON can provide a more natural deformation and preserve the original texture.

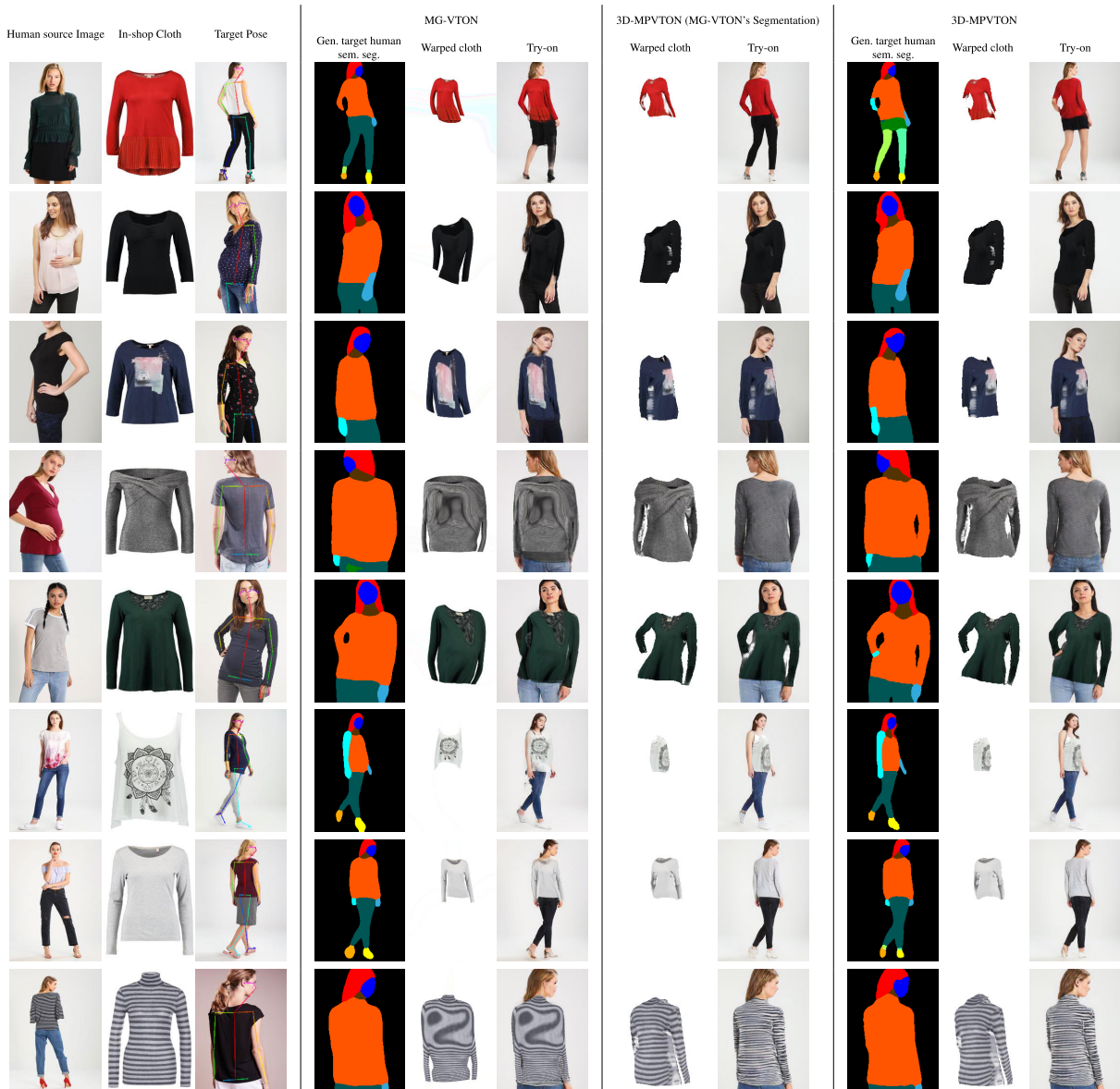


FIGURE 11. Qualitative comparison between our proposed method and MG-VTON on the test set with different clothes and different poses. From the left to right - Source human images, in-shop clothes, target new pose visualizations, human semantic segmentation generated by MG-VTON, 2D warped clothes generated by MG-VTON, the try-on results generated by MG-VTON, the 3D reconstructed clothes and the try-on results from our method based on MG-VTON's generated human semantic segmentation, the generated target human semantic segmentation from the proposed 3D-MPVTON, the 3D reconstructed clothes and the try-on results from the proposed 3D-MPVTON.

For example, for the large target poses with self-occlusions, our method can hide the appropriate parts of the clothing in side-view target poses, such as the right sleeve (second and eighth rows), or left sleeve (third row), or half-portions of the clothing texture (sixth row). Also, the first row shows that 3D-MPVTON generates the lower-body parts of target human semantic segmentation with legs and skirt based on the human source image, while MGVTON can only generate lower-body parts as pants. With the back-view awareness strategy, 3D-MPVTON's try-on results are significantly more realistic (the first, fourth, seventh, and eighth rows).

3) USER STUDY RESULT

A human perceptual evaluation study was performed for the try-on results on the MPV dataset [19]. We invited 109 participants to vote for the best generated try-on output from the results of the compared methods including ours. Each participant was presented 100 randomly selected samples from (the different clothing and pose) MPV testing set together with the human source image, new in-shop clothing, and a new pose, and asked to choose the most realistic try-on result with no time constraints.

Based on the participants' selections, 3D-MPVTON results were chosen 59.9% of the time, compared to MG-VTON.

TABLE 3. User study results on MPV [19] dataset. The study is conducted on the random tuples of new in-shop clothes and new target pose from the test dataset. The number and the percentage values indicate the preferences in each method in total, each clothing category, and user's gender.

Cloth type	Gender	MG-VTON [19]		3D_MPVTON	
		Num. of choices	Percentage	Num. of choices	Percentage
All	All	4368	40.1	6532	59.9
	Female	1949	39.0	3051	61.0
	Male	2419	41.0	3081	59.0
Long Sleeve	All	1058	36.7	1823	63.3
	Female	467	35.6	845	64.4
	Male	591	37.7	978	62.3
Short Sleeve	All	1760	41.5	2480	58.5
	Female	776	39.3	1199	60.7
	Male	984	43.4	1281	56.6
Sleeveless	All	1133	41.1	1624	58.9
	Female	519	41.3	738	58.7
	Male	614	40.9	886	59.1
Pregnennt Clothes	All	417	40.8	605	59.2
	Female	187	41.0	269	59.0
	Male	230	40.6	336	59.4

To avoid dataset distribution bias, we categorized the test dataset based on shapes, such as short sleeves, long sleeves, sleeveless, and pregnancy clothes. By examining the difference in each category, we noticed that our methods were selected more frequently (63.3%) for long-sleeve clothing cases.

We and the user study participants did not notice the identity loss, but the customer himself or herself would be more sensitive to the identity change. Later we could replace this step with a better human pose transfer and generation method.

VI. CONCLUSION

In this paper, we proposed 3D-MPVTON that utilized the 3D clothing reconstruction method in CloTH-VTON [5] for multiple-pose virtual try-on (VTON) scenarios. The results show that a hybrid approach based on the 3D clothing model reconstruction in CloTH-VTON can generate more natural clothing deformation results for multi-posed VTON applications. In comparison with 3D-MPVTON, 2D warping methods in the earlier works could not accommodate the kinds of clothing deformation needed for arbitrary target poses. Furthermore, we showed the importance of target segmentation accuracy and clothing texture mapping to 3D clothing model and developed novel training method and texture mapping stages, which are crucial for achieving high quality VTON results. The experiments with the MPV [19] dataset demonstrated that 3D-MPVTON can generate significantly more realistically deformed clothing with various poses than MG-VTON [19] and CloTH-VTON+ [6].

Similar to Cloth-VTON+ in the fixed pose scenario, the proposed 3D-MPVTON has limitations in the 3D clothing reconstruction step and the target pose and customer shape transfer step. Further study is required for application to the full range of diverse clothing styles, such as loose clothes and skirts, and for generating the natural folds and wrinkles of clothing draped on the human body.

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